

Complex Manifolds

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MA726

Abstract

MA722: Differential Geometry 2 was taught by [Professor Matiej Szczesny](#) in the spring 2026 semester (my junior year). Although not officially a topics course, it is treated by the department as a topics course and the officially listed course description is not accurate. This semester's topic is complex geometry, and we followed the book [\[Huy04\]](#).

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Chapter 1

Local theory

1.1 Holomorphic functions of several variables

Lecture 1: Syllabus day

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Book: *Complex Geometry* by Daniel Hugbrechts.

Definition 1.1.1. A complex manifold is a topological manifold locally modeled by open subsets of $\mathbb{C}^n$ such that the transition functions are biholomorphic (holomorphic and invertible).

Holomorphic functions are extremely rigid. For example, they are necessarily analytic (stronger than C^∞ – has convergent power series). These are very special manifolds and we should be able to say a lot about them. However, we cannot, for example, have partitions of unity or compactly supported holomorphic functions, so the study of them is very different from that of real manifolds, and we will use many tools that show up in algebraic geometry.

Recall: let $f : U \subseteq \mathbb{C} \rightarrow \mathbb{C}$. Then f is holomorphic if any of the following equivalent characterizations hold.

1. f satisfies the Cauchy-Riemann (C-R) equations: Let $z = x + iy$ and $f(z) = u(x, y) + iv(x, y)$. Then $\partial_x u = \partial_y v$ and $\partial_y u = -\partial_x v$.
2. $dfJ = Jdf$ for $J \in \text{Hom}_{\mathbb{C}}(\mathbb{C}, \mathbb{C})$ the complex structure $x \mapsto ix$ (easy verification, multiply matrices in both directions and recover C-R eqns).
3. Another way of saying (2) is df is $\mathbb{C}$ -linear.
4. Another formulation is defining $\partial_z = \frac{1}{2}(\partial_x - i\partial_y)$ and $\partial_{\bar{z}} = \frac{1}{2}(\partial_x + i\partial_y)$. Then we require $\partial_{\bar{z}}f = 0$. This follows by expanding it out in terms of u, v, x, y , and recovering C-R.
5. Near all $c \in U$, there exists an ε -ball $B(\varepsilon, c)$ centered at c where f has a convergent power series expansion.

We also recall some relevant theorems from single variable complex analysis.

Proposition 1.1.2 (Maximum principle). *Let $U \subseteq \mathbb{C}$ be open and connected. If $f : U \rightarrow \mathbb{C}$ is holomorphic and non-constant, then $|f|$ has no local maximum in U . If U is bounded and f can be extended to a continuous function $f : \bar{U} \rightarrow \mathbb{C}$, then $|f|$ takes its maximal values on the boundary ∂U .*

Proposition 1.1.3 (Identity theorem). *If $f, g : U \rightarrow \mathbb{C}$ are holomorphic on a connected open subset $U \subseteq \mathbb{C}$ such that for a non-empty open subset $V \subseteq U$, the restrictions $f|_V = g|_V$ agree, then $f = g$.*

Proposition 1.1.4 (Riemann extension theorem). *Let $f : B_\varepsilon(z_0) \setminus \{z_0\} \rightarrow \mathbb{C}$ be a bounded holomorphic function. Then f can be extended to a holomorphic function $f : B_\varepsilon(0) \rightarrow \mathbb{C}$.*

Proposition 1.1.5 (Riemann mapping theorem). *Let $U \subseteq \mathbb{C}$ be a simply connected proper open subset. Then U is biholomorphic to the unit ball $B_1(0)$, i.e. there exists a bijective holomorphic map $f : U \rightarrow B_1(0)$ such that its inverse is also holomorphic.*

Proposition 1.1.6 (Liouville). *Every bounded holomorphic function $f : \mathbb{C} \rightarrow \mathbb{C}$ is constant. In particular, there is no biholomorphic map between $\mathbb{C}$ and a ball $B_\varepsilon(0)$ with $\varepsilon < \infty$. This is a striking difference to the real situation and will cause a different concept of locality for complex manifolds than the one we are used to from real differential geometry.*

Proposition 1.1.7 (Residue theorem). *Let $f : B_\varepsilon(0) \setminus \{0\} \rightarrow \mathbb{C}$ be a holomorphic function. Then f can be expanded in a Laurent series*

$$f(z) = \sum_{n=-\infty}^{\infty} a_n z^n$$

and the coefficient a_{-1} is given by the residue formula

$$a_{-1} = \frac{1}{2\pi i} \int_{|z|=\varepsilon/2} f(z) dz.$$

The residue theorem is usually applied to more general situations where the function f has several isolated singularities in a connected open subset and the integral is taken over a closed contractible path surrounding the singularities.

We generalize this to several complex variables. We will not get into why this makes sense beyond contrasting with our previous characterizations since this subject is quite different than complex analysis in 1 variable.

We have notation as follows. $z \in \mathbb{C}^n$ is z_j with each component written as $z_j = x_j + iy_j$. We will often write i instead of j and assume it does not get confused with the complex unit. We will sometimes use $z^J = z_i^{j_k}$. A basis for the topology is given by polydisks

$$\Delta(\varepsilon, c) = \{z \in \mathbb{C}^n \mid |z_j - c_j| < \varepsilon_j \forall j = 1, \dots, n\}.$$

Polydisks in $\mathbb{C}^n$ are like real $2n$ -dimensional rectangles.

Theorem 1.1.8. *Let $U \subseteq \mathbb{C}^n$ open and $f \in C^\infty(U, \mathbb{C})$ (although assuming C^1 suffices). The following are equivalent.*

1. *f satisfies the C-R equations in each z_i : $\partial_{x_i} u = \partial_{y_i} v$ and $\partial_{x_i} v = -\partial_{y_i} u$;*
2. *f is separably holomorphic in each variable z_j , i.e. fixing all but one variable is holomorphic;*
3. *Given $c \in U$, there exists an ε -polydisk $\Delta(\varepsilon, c)$ such that $f(z)$ has a powerseries expansion in the polydisk $\Delta(\varepsilon, c)$*

$$f(z) = \sum_{|J|=0}^{\infty} a_J (z - c)^J.$$

Sketch of proof. Proof of (1) $\iff$ (2) is simple via 1-variable complex analysis. Power series converge on their interior, so they converge in any variable, thus (3) $\implies$ (2). The only work required to prove this is (2) $\implies$ (3). This follows from the multivariate Cauchy Integral Formula (CIF). $\blacksquare$

Definition 1.1.9. A function $f : U \subseteq \mathbb{C}^n \rightarrow \mathbb{C}$ is *holomorphic* if it satisfies any of the criteria of theorem 1.1.8.

Theorem 1.1.10 (Cauchy integration formula, [Huy04] 1.1.2). *Let f be holomorphic on a polydisk $D = \prod_j D_j$ with D_j an open disk, and continuous on its closure. Then for $z \in D$,*

$$f(z) = \frac{1}{(2\pi i)^n} \int \cdots \int_{\partial D_1 \times \cdots \times \partial D_n} \frac{f(\zeta_1, \dots, \zeta_n)}{\prod_j (\zeta_j - z_j)} d\zeta_1 \cdots d\zeta_n.$$

Many 1-variable theorems generalize. For example, the maximum modulus principle and the identity theorem. If f is a nonconstant holomorphic function, then its modulus does not attain a maximum on the interior of its domain. If f and g are analytic on a domain D and $f = g$ on some $S \subseteq D$ with an accumulation point, then $f = g$ on D .

Lemma 1.1.11 ([Huy04] 1.1.3). *Let $U \subseteq \mathbb{C}^n$ be open, and let $V \subseteq \mathbb{C}$ be an open neighborhood of the boundary of an ε -sphere $\partial B_\varepsilon(0)$. Suppose $f : U \times V \rightarrow \mathbb{C}$ is holomorphic. Then the function*

$$g(z_1, \dots, z_n) = \int_{|\zeta|=\varepsilon} f(\zeta, z_1, \dots, z_n) d\zeta$$

is holomorphic on U .

Sketch of proof. Differentiate under the integral sign; this is valid by compactness $|\zeta| = \varepsilon$, or alternatively the dominated convergence theorem. Thus

$$\frac{\partial g}{\partial \bar{z}_i} = \int_{|\zeta|=\varepsilon} \frac{\partial f}{\partial \bar{z}_i}(\zeta, z_1, \dots, z_n) d\zeta.$$

Since f is holomorphic, appealing to theorem 1.1.8 (2) the interior of the integral vanishes, and g is holomorphic also by theorem 1.1.8 (2). $\blacksquare$

More review.

Definition 1.1.12. Let $(X, \mathcal{T})$ be a topological space and $\text{Open}(X)$ the poset category $\mathcal{T}$. Then a *presheaf* with values in a *concrete* category $\mathcal{C}$ is a functor $\mathcal{F} : \text{Open}(X)^{\text{op}} \rightarrow \mathcal{C}$.

Often we take $\mathcal{C} = \text{Set}, \text{Ab}, R\text{-Mod}$, or another concrete abelian category $\mathcal{A}$. If $U \subseteq X$ is open, we call elements of $\mathcal{F}(U)$ *sections* of $\mathcal{F}$ over U . If $U \subseteq V$ are open in X and $s \in \mathcal{F}(V)$, then we write $s|_U$ for the image of s in the induced map $\mathcal{F}(V) \rightarrow \mathcal{F}(U)$.

Definition 1.1.13. A *sheaf* is a presheaf $\mathcal{F}$ that satisfies the following axioms.

- (Locality) If $U \subseteq X$ is open and $\{U_i \subseteq U\}_{i \in I}$ is an open cover of U , and $s, t \in \mathcal{F}(U)$ have $s|_{U_i} = t|_{U_i}$ for all i , then $s = t$.
- (Gluing) Let $U \subseteq X$ be open, $\{U_i \subseteq U\}_{i \in I}$ an open cover of U and $\{s_i \in \mathcal{F}(U_i)\}_{i \in I}$ a collection of sections. If $s_i|_{U_i \cap U_j} = s_j|_{U_i \cap U_j}$ for all $i, j \in I$, then there is some $s \in \mathcal{F}(U)$ such that $s_i = s|_{U_i}$ for all i .

Locality and gluing can be reinterpreted in terms of exactness of a sequence, and even more generally this can be rephrased as a certain diagram being an equalizer diagram. We will not need non-concrete categories in this case, so we ignore this generalization.

Notation 1.1.14. Given a space X , we define the *category of $\mathcal{C}$ -valued presheaves* as $\mathcal{Pshf}(X, \mathcal{C}) := \mathcal{C}^{\mathcal{O}pen(X)^{op}}$. We define the category $\mathcal{Shf}(X, \mathcal{C})$ of $\mathcal{C}$ -valued sheaves on X as the full subcategory of $\mathcal{Pshf}(X, \mathcal{C})$ spanned by sheaves. This means that a morphism of (pre)sheaves are natural transformations.

Definition 1.1.15. Let $\mathcal{F}$ be a (pre)sheaf of a space X on a category $\mathcal{C}$. The *stalk* $\mathcal{F}_p$ of $\mathcal{F}$ at a point $p \in X$ is the quotient

$$\{(s, U) \mid U \in \mathcal{O}pen(X), p \in U, s \in \mathcal{F}(U)\} / \sim$$

where $\sim$ is defined by $(s, U) \sim (t, V)$ iff there is an open set $W \subseteq U, V$ with $p \in W$ such that $s|_W = t|_W$. A *germ* of $\mathcal{F}$ at p is an element of the stalk of $\mathcal{F}$ at p . An equivalent characterization of the stalk is as follows. Let $\mathcal{U}_p \subseteq \mathcal{O}pen(X)$ be the full subcategory spanned by open sets containing p . Then

$$\mathcal{F}_p := \text{colim } \mathcal{F}|_{\mathcal{U}_p}.$$

Proposition 1.1.16. *Let $\mathcal{A}$ be an abelian category and X a space. Then $\mathcal{Pshf}(X, \mathcal{A})$ is abelian.*

Proof. By [Mac98], limits in functor categories exist and are computed pointwise. All statements that remain to be shown are very straightforward. ■

Corollary 1.1.17. *Let $\mathcal{A}$ be an abelian category and X a space. Then $\mathcal{Shf}(X, \mathcal{A})$ is abelian.*

Proof. It suffices to show sheaves are closed under taking (co)limits, which is trivial. ■

Example 1.1.18 ([Huy04] 1.1.14). Let $\mathcal{O}_{\mathbb{C}^n}$ denote the sheaf of holomorphic functions, where the sections of $\mathcal{O}_{\mathbb{C}^n}$ over $U \subseteq \mathbb{C}^n$ are all holomorphic functions $f : U \rightarrow \mathbb{C}$, and for each $U \subseteq V$ the restriction morphism $\mathcal{O}_{\mathbb{C}^n}(V) \rightarrow \mathcal{O}_{\mathbb{C}^n}(U)$ is precisely function restriction, meaning we map $f \in \mathcal{O}_{\mathbb{C}^n}(V)$ as $f \mapsto f|_U$. We write the stalk of $\mathcal{O}_{\mathbb{C}^n}$ at $z \in \mathbb{C}^n$ as $\mathcal{O}_{\mathbb{C}^n, z}$.

Lecture 2: Multivariate holomorphic functions and Weierstrass polynomials

Thu 22 Jan 2026 12:24

Theorem 1.1.19 (Hartog, [Huy04] 1.1.4). *Let there exist nested polydisks $\Delta(\varepsilon') \subseteq \Delta(\varepsilon)$ in $\mathbb{C}^n$ for $n \geq 2$, which we can WLOG take to be centered at 0. Let $U := \Delta(\varepsilon) \setminus \Delta(\varepsilon')$ and let $f : U \rightarrow \mathbb{C}$ be holomorphic. Then f extends to $\Delta(\varepsilon)$.*

Proof. Recall the following for 1-variable functions: if $f(z)$ is analytic in an annulus centered at 0, then $f(z)$ can be written

$$f(z) = \sum_{n \in \mathbb{Z}} a_n z^n, \quad a_n = \frac{1}{2\pi i} \oint_C f(\xi) \xi^{-n-1} d\xi$$

for any contour C with winding number 1 around the annulus (aka anything in the homotopy class; there are no singularities in the annulus by holomorphy).

We prove it in 2 variables. With assumptions and notation as stated in the problem, we can do a Laurant expansion in each z_i with the rest fixed. Hence

$$f(z) = \sum_{n \in \mathbb{Z}} a_n(z_1) z_1^n, \quad a_n(z_1) = \frac{1}{2\pi i} \oint_C f(z_1, \xi) \xi^{-n-1} d\xi.$$

Applying lemma 1.1.11 allows us to realize this as a holomorphic function in z_2 in the entire disk, hence $a_n(z_1) = 0$ for $n < 0$ and $\varepsilon'_1 < |z_1| < \varepsilon_1$. By the 1-variable identity principle, this implies $a_n(z_1) = 0$ for $n < 0$ in the entire disk. Thus only non-negative powers of z_2 occur in the expansion. The same logic applies for z_1 . Hence f extends via power series to the entire polydisk, and is particular holomorphic on $\Delta(\varepsilon)$. It is clear how an inductive proof can now proceed, or how this logic can prove $n > 1$ dimensions automatically. ■

Remark 1.1.20. This implies that in 2 or more variables, singularities are never isolated, nor are bounded. They look more like long codimension 1 hypersurfaces, in particular branched covers of $\mathbb{C}^{n-1}$.

Remark 1.1.21. This is an analytic analogue of the fact from AG that $\mathbb{A}^n \setminus \{0\}$, $n \geq 2$ is not affine.

We now explore precisely what singularities of multivariate complex variables look like. By passing to the reciprocal we can turn singularities into zeroes, so we start by trying to understand zero sets of analytic functions in $\mathbb{C}^n$.

Definition 1.1.22. A *Weierstrass polynomial* in z_1 is a polynomial of the form

$$z_1^d + \alpha_1(z_2, \dots, z_n) z_1^{d-1} + \alpha_2(z_2, \dots, z_n) z_1^{d-2} + \dots + \alpha_d(z_2, \dots, z_n)$$

where each α_i is a holomorphic function in some small disk in $\mathbb{C}^{n-1}$ about the origin, and they vanish at $0 \in \mathbb{C}^{n-1}$.

Proposition 1.1.23 (Weierstrass perparation, [Huy04] 1.1.6). *Let $f : \Delta(\varepsilon, 0) \rightarrow \mathbb{C}$ be holomorphic with $f(0) = 0$ and $f(z_1, 0, \dots, 0)$ not identically zero. Then in a polydisk $\Delta(\varepsilon', 0)$ with $\varepsilon' < \varepsilon$ we can write f as $p \cdot u$, where p is a Weierstrass polynomial and $u(0) \neq 0$ is a unit in the ring of germs at 0: $u \in \mathcal{O}_{\mathbb{C}^n, 0}$. In particular, p contains all the information of the zero set of f .*

I omit the proof in exchange for intuition for the zero set of f . Take f a function on $\mathbb{C}^2$. Fix a variable z_2 ; the Weierstrass polynomial is now a polynomial of degree $d - 1 = 3$ in z_1 which has $d - 1 = 3$ roots in the relevant slice of $\mathbb{C}^2$. Now letting z_2 vary shows that there are $d - 1 = 3$ roots in each slice which vary smoothly, yielding 3 lines (counting multiplicity) of singularities.

Theorem 1.1.24 (Weierstrass division). *Under the assumptions of proposition 1.1.23, let $p(z_1, \dots, z_n)$ be a Weierstrass polynomial in z_1 of degree d . In a sufficiently small $\Delta(\varepsilon, 0)$ we can write $f = q \cdot p + s$, where q is holomorphic in $\Delta(\varepsilon, 0)$ and $s \in \mathcal{O}_{\mathbb{C}^{n-1}}[z_1]$ is a polynomial in z_1 with coefficients in $\mathcal{O}_{\mathbb{C}^{n-1}}$ evaluated at a polydisk sufficiently close to 0, and is of degree less than d in z_1 . Note that s need not be a Weierstrass polynomial.*

Now we consider algebraic properties of the ring of germs of $\mathcal{O}_{\mathbb{C}^n, 0}$. This can be identified with the ring $\mathbb{C}\{z_1, \dots, z_n\}$ of convergent power series in a sufficiently small polydisk in all n variables by passing to the power series representation of holomorphic functions.

Remark 1.1.25. This is a power series because $\mathbb{C}\{z_1, \dots, z_n\}$ is obviously an integral domain.

Remark 1.1.26. $\mathcal{O}_{\mathbb{C}^n,0}$ is a *local ring*, meaning there is a unique maximal ideal. This is given by functions that vanish at the origin. To check this, we need only show that it is a unit. This indeed follows since it must have a multiplicative inverse.

Proposition 1.1.27. $\mathcal{O}_{\mathbb{C}^n,0}$ is a UFD.

Proof. We induct on n . Take $n = 1$. The only irreducible element up to units is $f(z) = z$, and all holomorphic functions are analytic. Assume the inductive hypothesis for $n - 1$. Recall Gauss' lemma implies that if R is a UFD, then $R[x]$ is a UFD. Take $f \in \mathcal{O}_{\mathbb{C}^n,0}$ a non-unit, meaning $f(0) = 0$. By proposition 1.1.23, $f = p \cdot u$ for u a unit in $\mathcal{O}_{\mathbb{C}^n,0}$ and p a Weierstrass polynomial, which lives in $p \in \mathcal{O}_{\mathbb{C}^{n-1},0}[z_1]$. By induction and Gauss' lemma, $\mathcal{O}_{\mathbb{C}^{n-1},0}[z_1]$ is a UFD, so factor p as irreducible polynomials $p = p_1 \cdots p_r$. We claim p_i are all Weierstrass. I omit the proof. We now need to show the p_i s are irreducible in $\mathcal{O}_{\mathbb{C}^n,0}$, since we are working in a proper subring $\mathcal{O}_{\mathbb{C}^{n-1},0} \subsetneq \mathcal{O}_{\mathbb{C}^n,0}$.

Suppose for contradiction that $p_j = a \cdot b$ in $\mathcal{O}_{\mathbb{C}^n,0}$, meaning one of these must be a unit. By proposition 1.1.23 on each of a, b , we obtain $a = q_1 u_q$ and $b = q_2 u_2$. Then $p_j = q_1 u_1 q_2 u_2$. We claim $q_1 q_2$ is also a Weierstrass polynomial. We then appeal to the uniqueness part of proposition 1.1.23 which we did not state but it says the polynomial is unique. Hence $u_1 u_2 = 1$, and thus p_j is a product of Weierstrass polynomials. Thus contradicts irreducibility of p_j . ■

Lecture 3: a

Tue 27 Jan 2026 12:29

Office hours are Wednesday 1-3pm, cds431. MA822 will probably move to psy b33.

Proposition 1.1.28. $\mathcal{O}_{\mathbb{C}^n,0}$ is Noetherian.

Proof. We induct on n . At $n = 1$ the statement is trivial apparently. Assume the inductive hypothesis for $n - 1$. Recall that the Hilbert basis theorem says that N Noetherian implies $N[x]$ Noetherian. We apply this here to obtain $\mathcal{O}_{\mathbb{C}^{n-1},0}[z_1]$ is Noetherian. Let $I \subseteq \mathcal{O}_{\mathbb{C}^n,0}$ be a nonzero proper ideal. Then it contains some nonzero nonunit $f \in I$, so $f(0) = 0$. By proposition 1.1.23, $f = p \cdot u$ for p a Weierstrass polynomial and u a unit. Then $f \in I$ implies $f \cdot u^{-1} = p \in I$. Consider $I' := I \cap \mathcal{O}_{\mathbb{C}^{n-1},0}$. Then since $\mathcal{O}_{\mathbb{C}^{n-1},0}$ is Noetherian, $I' = \langle q_1, \dots, q_r \rangle$. We claim $I = \langle p, q_1, \dots, q_r \rangle$. Choose $g \in I$. Then theorem 1.1.24 divides g as $ap + s$ for $s \in \mathcal{O}_{\mathbb{C}^{n-1},0}$ of degree less than p and a holomorphic. Since $g, p \in I$, we have $s \in I$. Thus $s \in I'$, and thus s is a linear combination in q_i . Thus g is a linear combination in p and the q_i s, concluding the proof. ■

Remark 1.1.29. We do not need to only consider rings of germs at the origin, since up to translation any point can be mapped to the origin.

Definition 1.1.30. Let $U \subseteq \mathbb{C}^n$ be open. An *analytic subvariety* of U is a subset $V \subseteq U$ such that for every $z_0 \in V$, there exists a collection of holomorphic functions $\{f_i : U \rightarrow \mathbb{C}\}_{1 \leq i \leq r}$ such that z_0 is in the intersection of the vanishing loci of the f_i s.

We can then discuss germs of analytic varieties, and obtain a nice language similar to that of algebraic geometry for $\mathbb{C}^n$. We will not continue down this path, and will instead begin our discussion of complex geometry.

Definition 1.1.31. Let $U \subseteq \mathbb{C}^n$ and $V \subseteq \mathbb{C}^m$ open. Then $f : U \rightarrow V$ is holomorphic iff all f_j s are holomorphic. We say f is biholomorphic if it has a holomorphic inverse.

Remark 1.1.32. With notation as in definition 1.1.31, if f is a biholomorphism, then $n = m$ is forced. We do not prove this since it is not quite trivial.

Notation 1.1.33. We will generally write coordinates in the domain as $z_i = x_i + iy_i$ and coordinates in the codomain as $w_i = u_i + iv_i$.

Identifying $\mathbb{C}^n \cong \mathbb{R}^{2n}$, the derivative takes the form

$$Df = \begin{bmatrix} \frac{\partial u^i}{\partial x_j} & \frac{\partial u^i}{\partial y_j} \\ \frac{\partial v^i}{\partial x_j} & \frac{\partial v^i}{\partial y_j} \end{bmatrix}.$$

This is $\mathbb{C}$ -linear $T_a\mathbb{C}^n \rightarrow T_{f(a)}\mathbb{C}^m$, which are complex vector spaces now defined as $C^\infty(\mathbb{C}^n)$. In this case, Df is represented by the frames ∂_{x_i} and ∂_{y_i} by

$$Df_i^j = \frac{\partial f^j}{\partial z_i}.$$

For example, take $U, V \subseteq \mathbb{C}$. The fact that f is holomorphic and Df viewed as $\mathbb{R}^2 \rightarrow \mathbb{R}^2$ is $\mathbb{C}$ -linear means that for J the complex structure

$$J = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix},$$

we have $J \circ Df = Df \circ J$. Expanding this out yields

$$Df(\partial_x) = \frac{\partial f}{\partial z} \partial_u.$$

It is convenient to work with complex tangent vectors.

Warning 1.1.34. The vectors ∂_{z_j} and $\partial_{\bar{z}_j}$ *do not live* in $T_a\mathbb{R}^{2n}$. This is because $\partial_{z_j} = \frac{1}{2}(\partial_{x_j} + i\partial_{y_j})$, and the number i doesn't make sense in $T_a\mathbb{R}^{2n}$. Thus we need to tensor with $\mathbb{C}$ over $\mathbb{R}$. Thus we usually consider $T_a\mathbb{C}^n \otimes_{\mathbb{R}} \mathbb{C}$.

Claim 1.1.35. *With respect to the frames ∂_{z_j} and $\partial_{\bar{z}_k}$, $Df : T_a\mathbb{C}^n \rightarrow T_{f(a)}\mathbb{C}^m$ extends to a linear transformation*

$$Df : T_a\mathbb{C}^n \otimes_{\mathbb{R}} \mathbb{C} \rightarrow T_a\mathbb{C}^n \otimes_{\mathbb{R}} \mathbb{C}.$$

Moreover it is of the form

$$Df = \begin{bmatrix} (\partial_z f) & (\partial_{\bar{z}} f) \\ (\partial_z \bar{f}) & (\partial_{\bar{z}} \bar{f}) \end{bmatrix}.$$

Thus if it is holomorphic,

$$Df = \begin{bmatrix} (\partial_z f) & 0 \\ 0 & (\partial_{\bar{z}} \bar{f}) \end{bmatrix} = \begin{bmatrix} (\partial_z f) & 0 \\ 0 & \overline{(\partial_z f)} \end{bmatrix}.$$

Let's see this when $m = n = 1$. Then

$$Df(\partial_z) = \begin{pmatrix} \partial_x u & \partial_y u \\ \partial_x v & \partial_y v \end{pmatrix} \begin{pmatrix} \frac{1}{2} \\ \frac{i}{2} \end{pmatrix} = \frac{1}{2} \begin{pmatrix} \partial_x u \\ \partial_x v \end{pmatrix} - \frac{i}{2} \begin{pmatrix} \partial_y u \\ \partial_y v \end{pmatrix} = \frac{1}{2} \begin{pmatrix} \partial_x u - i\partial_y u \\ \partial_x v - i\partial_y v \end{pmatrix}.$$

We claim this is

$$\frac{\partial f}{\partial z} \frac{\partial}{\partial w} + \frac{\partial \bar{f}}{\partial z} \frac{\partial}{\partial \bar{w}}.$$

This can be verified by brute force. We remark that since $T_a\mathbb{C}^n \otimes_{\mathbb{R}} \mathbb{C}$ is basically just $T_a\mathbb{C}^n \oplus iT_a\mathbb{C}^n$ and since the Jacobians of holomorphic functions are half zero, they basically reduce to maps $T_a\mathbb{C}^n \rightarrow T_a\mathbb{C}^n$ in the way we would conventionally think, and the determinant agrees with their real Jacobian.

Lecture 4: sldfkskm

Thu 29 Jan 2026 12:28

Theorem 1.1.36 (Implicit function, real version). *If $f : U \subseteq \mathbb{R}^n \rightarrow V \subseteq \mathbb{R}^m$, U, V open, and $Df_a : T_a \mathbb{R}^n \rightarrow T_{f(a)} \mathbb{R}^m$ is surjective for all $a \in f^{-1}(f(a))$, then if $f^{-1}(f(a))$ is the graph of a smooth function, then $f^{-1}(f(a))$ is a manifold.*

Theorem 1.1.37 (Lazarfeld 1.19; Inverse and implicit function). *Let $f : U \subseteq \mathbb{C}^n \rightarrow V \subseteq \mathbb{C}^m$ be holomorphic with a Jacobian that is surjective (at a iff neighborhood of a).*

1. *If $n = m$, then there exist neighborhoods $U' \ni a$ and $V' \ni f(a)$ such that $f|_{U'}$ is a biholomorphism to V' .*
2. *If $n > m$ then $f^{-1}(f(a))$ is locally the graph of a holomorphic map.*

Proof. Existence of C^∞ solutions to (1,2) follow by the smooth versions of inverse and implicit function theorems, so we prove only holomorphy. ■

Definition 1.1.38 (Laz 2.1). Let X be a smooth manifold.

1. A holomorphic atlas $\mathcal{A} = \{(U_i, \varphi_i)\}_{i \in I}$ consists of an open cover $\{U_i\}_{i \in I}$ together with diffeomorphisms $\varphi : U_i \cong V_i \subseteq \mathbb{C}^n$, n fixed, V_i open, such that the transition functions $\varphi_j \circ \varphi_i^{-1}$ are holomorphic for all $i, j \in I$.
2. Two holomorphic atlases are *compatible* if their union is a holomorphic atlas.
3. A complex manifold of dimension n is a smooth $2n$ -dimensional manifold together with an equivalence class of compatible holomorphic atlases (basically the same as a maximal atlas).

Definition 1.1.39 (Laz 2.2). Let X be a complex manifold.

1. A function $f : X \rightarrow \mathbb{C}$ is *holomorphic* if $F \circ \varphi_i^{-1} : \varphi(U_i) \subseteq \mathbb{C}^n \rightarrow \mathbb{C}$ is holomorphic for every i .
2. If Y is also a complex manifold, then $f : X \rightarrow Y$ is holomorphic if (writing $\{V_j, \psi_j\}_{j \in J}$ for the atlas on Y), for all i, j ,

$$\psi_j \circ F \circ \varphi_i^{-1} : \varphi(U_i) \rightarrow \psi(V_j)$$

is holomorphic $\mathbb{C}^n \rightarrow \mathbb{C}^m$.

Example 1.1.40. All open subsets of complex manifolds are complex manifolds by restricting along the charts. A more interesting example is by viewing S^2 as the one point compactification of $\mathbb{R}^2$ via taking the north pole to be ∞ , and identifying $S^2 \setminus \{N\}$ with $\mathbb{R}^2$ via the stereographic projection. The transition map from the north and south charts is $z \mapsto 1/z$, which is holomorphic.

Now we do analytic hypersurfaces. Let $U \subseteq \mathbb{C}^{n+1}$ open and $f : U \rightarrow \mathbb{C}$ holomorphic. Take $Z(f)$ the zero locus of f inside of U . The implicit function theorem tells us when this is a manifold. Let for all $a \in Z(f)$, the vector $\partial_{z_i} f(a)$ be nonzero for some i . Then by the IFT, $Z(f)$ is locally given as the graph of a holomorphic map $g : (V \subseteq \mathbb{C}^n) \rightarrow Z(f)$; i.e. $Z(f) = \{(w, g(w))\}$. This is then a manifold.

It is important to be careful and ensure that spaces are bounded. For example, the zero locus of $z_1^2 + z_2^2 - 1$ is not bounded.

Lecture 5: Complex manifolds

Tue 03 Feb 2026 12:33

Example 1.1.41. If X, Y are complex manifolds, then $X \times Y$ is a complex manifold since it is a smooth manifold, and products of holomorphic functions are holomorphic by definition.

Let $\{f_i\}_{1 \leq i \leq k}$ holomorphic $X \rightarrow \mathbb{C}$, X a complex manifold. Then we can define the zero locus on *all* functions $Z(f_1, \dots, f_k) \subseteq X$ as

$$\{z \in X \mid f_i(z) = 0 \forall i\}.$$

If

$$J(f_1, \dots, f_k)_j^i = \frac{\partial f_i}{\partial z_j}$$

has full rank at all z , then the zero locus is a complex submanifold of X . We do this by taking the zero locus as a curve-like thingy and straightening it.

Another example is complex Lie groups. These are very simply Lie groups that are complex manifolds with multiplication and inversion holomorphic. Equivalently, it is a group object in $\mathbb{C}\text{plxMan}$. For example, consider $\text{GL}(n, \mathbb{C})$. This is equal to $M_{n \times n}(\mathbb{C}) \setminus Z(\det)$, and since the fiber of 0 is closed, $\text{GL}(n, \mathbb{C})$ is open in $\mathbb{C}^{2n}$. Since multiplication is a polynomial in each entry, it is holomorphic. Inversion is also holomorphic since it can, via Cramer's rule, be realized as a division of polynomials, which is a rational function in the entries in this case, and we are working in the nonvanishing locus of $\det$.

There are many subgroups of GL that are submanifolds. For example, $\text{SL}(n, \mathbb{C})$ the set of all matrices with determinant 1. This is the fiber of 1 under $\det$, it suffices to check that 1 is a regular value of $\det$, meaning the Jacobian is surjective on the tangent spaces on each element of the preimage. It suffices to show $d(\det)_A : T_A \text{GL}(n, \mathbb{C}) \rightarrow T_1 \mathbb{C}$ for all $A \in \text{SL}(n, \mathbb{C})$. We are able to work in the non-complexified tangent space, since that will be surjective precisely when the real Jacobian is surjective (follow directly from writing down the complexified Jacobian). To do this, we take a curve passing through A and show its derivative maps to a nonzero tangent vector. Let $\gamma(t) = (1+t)A$. Then

$$\det(\gamma(t)) = (1+t)^n \det A = (1+t)^n.$$

Thus

$$d(\det)_A(\gamma'(0)) = \left. \frac{d}{dt} \right|_0 \gamma(t) = n \neq 0.$$

Many other matrix Lie groups are defined analogously. However, *the unitary and special unitary groups are NOT complex Lie groups*. This follows by a fact that compact complex Lie groups have to be abelian. This follows by considering conjugation, which you can show is always constant. Then since U , SU are compact and nonabelian, the statement follows.

Another example is complex tori. Let $\Lambda \in \mathbb{C}^n$ be a lattice: a free abelian subgroup of the maximal rank $2n$. Hence this is simply the subgroup $\langle \lambda_i \rangle_{1 \leq i \leq 2n}$ for $\lambda_i \in \mathbb{C}^n$ linearly independent. Now define the group quotient $X_\Lambda := \mathbb{C}^n / \Lambda$. The quotient map is the universal cover of X_Λ , and moreover is a local diffeomorphism. Hence we obtain coordinates on X_Λ by taking a local inverse s on U of π , then defining coordinates on U via $s : U \rightarrow V \subseteq \mathbb{C}^n$. Coordinate transformations are translations – affine maps. Those are clearly holomorphic. We claim there is a diffeomorphism $X_\Lambda \cong (S^1)^{2n}$. NOTE that this is NOT an isomorphism of complex manifolds.

We can construct a matrix $A = (\lambda_1 \ \dots \ \lambda_{2n})$, which is $\mathbb{R}$ -linear and invertible since the vectors λ_i are linearly independent. Thus the inverse is $\lambda_i \mapsto e_i$. This then descends to $\mathbb{R}^{2n} / \Lambda \cong \mathbb{R}^{2n} / \langle e_i \rangle$. We commute the product and quotient as $\prod_{i=1}^{2n} \mathbb{R} / \mathbb{Z} \cong \prod_{i=1}^{2n} S^1$.

However, not all are biholomorphic. The issue is that the matrix need not be $\mathbb{C}$ -linear. We claim all holomorphic maps $X_\Lambda \rightarrow X_{\Lambda'}$ are induced by some $A \in M_{n \times n}(\mathbb{C})$. The universal cover of both of these is $\mathbb{C}^n$, so if $f : X_\Lambda \cong X_{\Lambda'}$, then there is a lift

$$\begin{array}{ccccc} & & & \mathbb{C}^n & \\ & & \nearrow F & \downarrow \pi & \\ \mathbb{C}^n & \xrightarrow{\pi} & X_\Lambda & \xrightarrow{f} & X_{\Lambda'} \end{array}$$

by precomposing X_Λ with its cover. Since F is holomorphic, consider $g(z) = F(z + \lambda_i) - F(z)$. We can show that $g(z) \in \Lambda'$ for all $z \in \mathbb{C}^n$. Note that Λ' is discrete and $\mathbb{C}^n$ is connected. Thus g is constant for all λ_i . We take the derivative to obtain that the holomorphic derivatives of the lift are periodic with respect to Λ . Hence the values are fully determined within one “box” determined by the lattice. Hence $\partial_{z_j} F$ is bounded and holomorphic, since it is determined on a compact bounded subset. Liouville’s theorem says bounded holomorphic functions on $\mathbb{C}$ are constant. Thus we are constant in each variable, and $\partial_{z_j} F$ is constant. Thus we must have F of the form

$$F(z) = \tilde{A}z + b,$$

$\tilde{A}$ an $n \times n$ matrix and $b \in \mathbb{C}^n$ a constant. Translating by $-b$ we can take $F(z) = \tilde{A}z$. Hence our map is (affine; 0 not always mapped to 0) $\mathbb{C}$ -linear.

Since lattices correspond to invertible $2n \times 2n$ matrices, over $\mathbb{R}$ the space of lattices can be defined with dimension $4n^2$. However $\mathrm{GL}(n, \mathbb{C})$ has $\mathbb{R}$ -dimension $2n^2$. Thus there are more lattices than these linear maps, and thus not all can be biholomorphic.

Final example for today are Hopf manifolds. Pick $\lambda \in \mathbb{R}$ with $|\lambda| < 1$. Then $\mathbb{Z}$ acts on $\mathbb{C} \setminus \{0\}$ by $m \cdot (z_1, \dots, z_n) = (\lambda^m z_1, \dots, \lambda^m z_n)$. The action is free and *properly discontinuous*, meaning (for an arbitrary G -space) for any open set U and any $g \in G$, then $g(U) \cap U$ is nonempty iff $g = e$. A Hopf manifold is $\mathbb{C}^n \setminus \{0\}$ quotiented by this action. Interestingly, as a complex manifold, this is isomorphic to $S^{2n-1} \times S^1$.

We will generally say *submanifold* to mean an embedded submanifold, and will specify when submanifolds are immersed explicitly.

Lecture 6: Complex furries

Thu 05 Feb 2026 12:28

Remark 1.1.42. I finally have good motivation for why complexifying the tangent space is tensoring with $\mathbb{C}$. Let $v \in T_p M$ and $z = x + iy \in \mathbb{C}$. Then $v \otimes z \in T_p^{\mathbb{C}} M$ is equivalent to, by bilinearity,

$$v \otimes (x + iy) = v \otimes x + v \otimes iy.$$

Of course by $\mathbb{R}$ -linearity of $\otimes_{\mathbb{R}}$, this is equal to

$$= x(v \otimes 1) + y(v \otimes i).$$

We can choose to write $v \otimes 1 = v$ and $v \otimes i = iv$, and this really does behave like normal multiplication since normal multiplication is also bilinear. Thus all elements of $T_p^{\mathbb{C}} M$ can be written of the form $u + iv$ for some $u, v \in T_p M$; in our example above, $u = xv$ and $v = yv$ (abusing notation very badly). This is required to have $\partial_z = \frac{1}{2}(\partial_x - i\partial_y)$ in the tangent space.

Proposition 1.1.43. *Let X be a compact complex connected submanifold, and $f : X \rightarrow \mathbb{C}$ a holomorphic function. Then f is constant.*

Proof. Easy to show f is constant on connected components by passing to coordinates. ■

Corollary 1.1.44. *If $X \subseteq \mathbb{C}^n$ is a compact complex connected submanifold, then X is a point.*

Proof. Take the holomorphic function

$$X \hookrightarrow \mathbb{C}^n \xrightarrow{\pi_i} \mathbb{C}.$$

This must be constant by definition 1.1.47. ■

Since looking for compact complex connected subsets of $\mathbb{C}^n$, we look to projective space for something like Whitney's theorem for embedding complex manifolds into $\mathbb{C}^n$. This still is not perfect, but is worth considering.

Recall that projective space $k\mathbb{P}^n$ over a field k is defined as the set of lines through the origin in $k^{n+1} \setminus \{0\}$, topologized in the obvious way:

$$k\mathbb{P}^n := \frac{k^{n+1} \setminus \{0\}}{\langle \lambda x \sim x \mid x \in k^{n+1}, \lambda \in k \setminus \{0\} \rangle}.$$

Given π the canonical projection and L any line in $\mathbb{C}^{n+1}$, we have $\pi^{-1}(L) \cong L \setminus \{0\} \cong \mathbb{C}^*$. We can then show that $\mathbb{C}^{n+1} \setminus 0$ is a locally trivial $\mathbb{C}^*$ -bundle over $\mathbb{CP}^n$. Recall that the atlas is formed by sets

$$U_i := \{[c_0 : \cdots : c_n] \in \mathbb{CP}^n \mid c_i \neq 0\}.$$

This is well-defined. The coordinates simply drop c_i , since we can always take it to be normalized at 1.

Remark 1.1.45. We work in the quotient topology on $\mathbb{CP}^n$, which is not the same as the usual topologies we would use in algebraic geometry. However many of the results translate between the two, due to some work by Serre.

We equip $\mathbb{C}^{n+1}$ with the Hermitian inner product $\langle z, w \rangle = \sum_i z_i \bar{w}_i$, the *opposite* of the physics convention.

The fibration

$$S^1 \hookrightarrow S^{n+1} \longrightarrow \mathbb{CP}^n$$

is called the Hopf fibration.

Complex projective space is a CW complex. We can define $\mathbb{CP}^1 := \mathbb{C}^1 \amalg \mathbb{C}^0$ with $\mathbb{C}^0$ the point at ∞ . If we take a chart $U_0 \subseteq \mathbb{CP}^2$, then $\mathbb{CP}^2 \setminus U_0$ is just the set $\{[0 : z_1 : z_2]\}$. Since $U_0 \cong \mathbb{C}^2$, we have $\mathbb{CP}^2 := \mathbb{C}^2 \amalg \mathbb{C}^1 \amalg \mathbb{C}^0$. We can induct to obtain

$$\mathbb{CP}^n := \coprod_{i=0}^n \mathbb{C}^i.$$

Hence $\mathbb{CP}^n$ admits this cell decomposition and is a CW complex with cells of even real dimension. Here we of course have not made the gluing relations explicit. Since cohomology of a CW complex comes by each level of the cells, and every other degree of cells is zero, the cohomology $H^\bullet(\mathbb{CP}^n; \mathbb{C}) \cong \mathbb{C}/0 \cong \mathbb{C}$. Oh and also $\mathbb{CP}^n \cong S^{n+1}/S^1$ a diffeomorphism.

Submanifolds (more generally, subvarieties) of $\mathbb{CP}^n$ admit descriptions as *homogeneous polynomials*. A polynomial f in $n+1$ variables is *homogeneous of degree d* if $f(\lambda x) = \lambda^d f(x)$, $x \in \mathbb{C}^{n+1}$. It follows that f is a sum of monomials of degree d .

Lecture 7: Complex projective space

Tue 10 Feb 2026 12:29

Notation 1.1.46. Given a manifold X and an open set $U \subseteq X$, we will now write $\mathcal{O}_X(U)$ for the set of holomorphic functions $U \rightarrow \mathbb{C}$. Recall by ?? that if X is compact, then $\mathcal{O}_X(X) \cong \mathbb{C}$.

Let f be a homogeneous polynomial of degree d . It makes sense to consider the vanishing locus $Z(f)$ as a subset of $\mathbb{CP}^n$ since if f vanishes on z , then it vanishes on $\{\lambda z\}_{\lambda \in \mathbb{C}} = [z]$.

Definition 1.1.47. Let $F_1, \dots, F_r$ be homogeneous polynomial of degree d in $\mathbb{C}[z_0, \dots, z_n]$. Then the zero locus is defined as

$$Z(F_1, \dots, F_r) = \{[z] \in \mathbb{CP}^n \mid F_i([z]) = 0 \forall i\}.$$

Subsets of this are called *projective varieties*.

Note that any complex (embedded) manifold of $\mathbb{CP}^n$ is compact.

Theorem 1.1.48 (Chow). *Any complex submanifold of $\mathbb{CP}^n$ is cut out by homogeneous polynomials. In particular, all complex submanifolds of $\mathbb{CP}^n$ are algebraic.*

Proposition 1.1.49. *Given F a homogeneous polynomial in $\mathbb{C}[z_0, \dots, z_n]$, suppose that $\partial_{z_i} F$ has no common zeroes in $\mathbb{CP}^n$ for all i . Then $Z(F)$ is a manifold.*

Proof. Let F be homogeneous of degree d . There is an identity known as *Euler's identity* which says

$$\sum_{i=0}^n z_i \frac{\partial F}{\partial z_i} = dF(z).$$

Here d is the degree, NOT the exterior derivative (it is my convention to write that as d). We prove this by differentiating $F(\lambda z) = \lambda^d F(z)$ on both sides with respect to λ and applying the chain rule, then setting $\lambda = 1$.

We want to verify that $Z(F) \cap U_i$ is a manifold. For simplicity take $i = 0$ without loss of generality. Take $[a] \in Z(F) \cap U_i$; we then may freely set $a_i = 1$. Using the identity,

$$\left. \frac{\partial F}{\partial z_0} + a_1 \frac{\partial F}{\partial z_1} + \dots + a_n \frac{\partial F}{\partial z_n} \right|_{(1, a_1, \dots, a_n)} = dF(1, a_1, \dots, a_n).$$

The right side vanishes. Hence $Z(F) \cap U_0 = Z(F(1, z_1, \dots, z_n))$. To check that this is a manifold using the implicit function theorem, we need a nonzero partial derivative, and thus the Jacobian will have rank 1. The hypothesis and non-zero-ness of $[a]$ guarantees this. ■

Let $X \subseteq \mathbb{CP}^n$ a manifold or projective variety, and $\{F_i\}_{0 \leq i \leq s} \subseteq \mathbb{C}[z_0, \dots, z_n]$ all homogeneous of the same degree d . We claim the map $f : X \rightarrow \mathbb{CP}^s$ defined by $[x] \mapsto [F_0(x) : \dots : F_s(x)]$ is well-defined. This follows by homogeneity of the F s, *except* for the fact that $f[x]$ needs to be nonzero everywhere. This happens if and only if $X \cap Z(F_0, \dots, F_s)$ is empty. We claim f is holomorphic. Let $x \in U_j$ and $f(x) \in U_k$, then in coordinates

$$f(x) = \left\{ \frac{F_i(z_0, \dots, \hat{z}_j, \dots, z_n)}{F_k(z)} \right\}_{0 \leq i \leq s}.$$

This is holomorphic. A fascinating result we will see later is that *all* holomorphic maps arise of this form.

Recall that $\mathrm{GL}(n+1, \mathbb{C})$ acts on $\mathbb{C}^{n+1}$, and this descends to $\mathbb{CP}^n$. However this is not a faithful action, since some matrices act trivially: anything that scales stuff. This is the center, so we can mod out by the center $\mathrm{GL}(n+1, \mathbb{C})/\mathbb{Z}(\mathrm{GL}(n+1, \mathbb{C}))$, and this acts faithfully. We claim that these are precisely the automorphisms of $\mathbb{CP}^n$. We may prove this later.

Let $X \subseteq \mathbb{CP}^r$ and $Y \subseteq \mathbb{CP}^s$ are manifolds. Any submanifold of projective space will be called a projective manifold. We know $X \times Y$ is a complex manifold, but we do not yet know if it is projective. We proceed with the following proposition; $X \times Y$ will then be projective by a simple corollary.

Proposition 1.1.50 (Segre's embedding). $\mathbb{CP}^r \times \mathbb{CP}^s$ is a submanifold of $\mathbb{CP}^{rs+r+s}$.

Proof. Think of $\mathbb{C}^{rs+r+s+1} = \mathbb{C}^{(r+1)(s+1)}$ as matrices of size $(r+1) \times (s+1)$ over $\mathbb{C}$. Then $\mathbb{CP}^{rs+r+s}$ is just $\mathbb{P}(M_{(r+1) \times (s+1)}(\mathbb{C}))$. Consider the map $\mu_{r,s} : \mathbb{CP}^r \times \mathbb{CP}^s \rightarrow \mathbb{P}(M_{(r+1) \times (s+1)}(\mathbb{C}))$ defined by

$$([a], [b]) \mapsto [c_{ij}], \quad c_{ij} = a_i b_j.$$

For example,

$$([a_0 : a_1], [b_0 : b_1]) \mapsto \left[\begin{pmatrix} a_0 b_0 & a_0 b_1 \\ a_1 b_0 & a_1 b_1 \end{pmatrix} \right],$$

where of course the class is defined by $c_{ij} \sim \lambda c_{ij}$. We claim the image of this map is precisely (classes of) all matrices of rank 1 (all matrices in a class of a matrix with rank 1 will have rank 1). The argument is messy so we do it for our 2 dimensional example. We can take a_0 out of the first row and a_1 out of the second row to get

$$\begin{pmatrix} b_0 & b_1 \\ b_0 & b_1 \end{pmatrix}.$$

This has at most rank 1. Since $a, b \neq 0$ it has at least rank 1. In this example, this is a submanifold since it is precisely the fiber of 0 under $\det$. ■

Lecture 8: Toric geometry

Thu 19 Feb 2026 12:22

Class taught by Lau today. Consider $\mathbb{C}^\times = \mathbb{C} \setminus \{0\}$. Topologically this is a cylinder. We have a map $\mathbb{C}^\times \rightarrow \mathbb{R}$ via $z \mapsto \log |z|$. Note that S^1 acts on $\mathbb{C}^\times$ by multiplication, and our map $z \mapsto \ln |z|$ is S^1 -equivariant. Now consider taking the dimension to n :

$$(\mathbb{C}^\times)^n \rightarrow \mathbb{R}^n, \quad \{z_i\}_{1 \leq i \leq n} \mapsto \{\ln |z_i|\}_{1 \leq i \leq n}.$$

Then the n -torus $T^n = (S^1)^n$ acts on this in the same way, and our map is T^n -equivariant.

We might want to allow this to be $\mathbb{C}$ instead of $\mathbb{C}^\times$, but $\ln 0 = -\infty$, so we need to take our map $\mathbb{C} \rightarrow \mathbb{R} \cup \{-\infty\}$. We then think of $\mathbb{C}$ as kind of like a cylinder that is infinite in one dimension and pinched off at 0 in the other, since each circle is identified down by taking $\ln$ of its modulus. We could then try something like

$$\mathbb{C}^n \times (\mathbb{C}^\times)^m \rightarrow (\mathbb{R} \cup \{-\infty\})^n \times \mathbb{R}^m$$

defined in the same T^{n+m} -equivariant way.

Consider $\mathbb{CP}^1$. We can apply our toric point of view here as well. Consider the map $\mathbb{CP}^1 \rightarrow \mathbb{R}$ in coordinates as follows. In U_0 , we map $[z_0 : z_1] = [1, z_1/z_0] \mapsto \log |z_1/z_0|$. We do the same for U_1 . We can do a similar construction for $\mathbb{CP}^2 \rightarrow \mathbb{R}^2$ by dividing out by the nonzero element of a chart and applying our construction on each coordinate. This allows us to visualize $\mathbb{CP}^n$ as polytopes and

stuff in $\mathbb{R}^n$ modulo a toric action. It also makes it much easier to read off T^n -equivariant morphisms of complex manifolds.

A *toric variety* is a n -variety X that admits a $(\mathbb{C}^\times)^n$ -action with an open dense orbit. We then take the T^n -subaction. Importantly, we can also reverse this process and obtain a manifold back by tating the dual cone of a cone and then just constructing back our manifold.

Lecture 9: Holomorphic vector bundles

Thu 26 Feb 2026 12:31

Consider a map $X \rightarrow \mathbb{CP}^m$ for $X \subseteq \mathbb{CP}^n$ a manifold. Such a map is given by a collection $\{p_i\}_{0 \leq i \leq m}$ of homogeneous polynomials of the same degree that do not simultaneously vanish on X . We will see that this can be described as follows: maps $X \rightarrow \mathbb{CP}^m$ are given by a line bundle $\mathcal{L} \rightarrow X$ together with a collection of holomorphic sections that don't simultaneously vanish. To understand this, we define holomorphic vector bundles.

There are two approaches to holomorphic vector bundles. We start with the *total space picture*, and show how this is equivalent to the *cocycle picture*.

Definition 1.1.51. Let X be a complex manifold. A holomorphic vector bundle E over X of rank k is a complex manifold E together with a holomorphic map $\pi : E \rightarrow X$ such that

- there exists an open cover $\{U_i\}_{i \in I}$ of X and biholomorphisms $\varphi_i : \pi^{-1}(U_i) \rightarrow U_i \times \mathbb{C}^k$ making the following diagram commute:

$$\begin{array}{ccc} \pi^{-1}(U_i) & \xrightarrow{\varphi_i} & U_i \times \mathbb{C}^k \\ & \searrow \pi \quad \swarrow \pi_i & \\ & U_i & \end{array}$$

- Write $U_{ij} = U_i \cap U_j$. Then the transition functions $\varphi_{ij} := \varphi_i \circ \varphi_j^{-1} : U_{ij} \cap \mathbb{C}^k \rightarrow U_{ij} \cap \mathbb{C}^k$ must be biholomorphic and of the form

$$\varphi_{ij}(x, v) = (x, g_{ij}(x)v)$$

for some $g_{ij} \in \text{GL}(k, \mathcal{O}_X(U_{ij}))$; i.e. g_{ij} is a biholomorphic map $U_{ij} \rightarrow \text{GL}(k, \mathbb{C})$.

Note that $g_{ij} = g_{ji}^{-1}$, and on a triple overlap U_{ijk} , we have $g_{ij} \cdot g_{jk} = g_{ik}$. This is called the *cocycle condition*, and the collection $\{g_{ij}\}_{i,j \in I}$ is called the *cocycle* of E .

Terminology 1.1.52. A holomorphic vector bundle of rank 1 is called a holomorphic *line bundle*.

Example 1.1.53. A simple example is the trivial bundle $X \times \mathbb{C}^k \rightarrow X$. For a more interesting example, consider $\mathcal{O}_{\mathbb{CP}^n}(-1)$. Given $[a] \in \mathbb{CP}^n$, write L for the line $\{([a], v) \mid v \in \mathbb{C}a\}$. We project by $p : L \rightarrow \mathbb{CP}^n$. We use the usual cover $\{U_i\}_{0 \leq i \leq n}$ where the i th homogeneous coordinate is nonzero in U_i , and its preimage $p^{-1}(U_i)$ is $U_i \times \{v \in \mathbb{C}^{n+1} \mid v \in [z] \in U_i\}$. We then write $v = t[z]$ for the representative of $[z]$ with $z_i = 1$, and map $([z], v) \mapsto ([z], t)$. This gives a map $p^{-1}(U_i) \cong U_i \times \mathbb{C}$. By construction the relevant diagram commutes. Since $\text{GL}(1, \mathcal{O}_X(U_{ij})) \cong \mathcal{O}_X(U_{ij})$, we compute the cocycle by finding a holomorphic function which is nonvanishing on the overlaps. Define

$$g_{ij} : U_{ij} \rightarrow \text{GL}(1, \mathbb{C}) = \mathbb{C}^*$$

as follows. Define

$$g_{ij} : U_{ij} \times \mathbb{C} \rightarrow U_{ij} \times \mathbb{C}$$

as follows. View the first U_{ij} as $\subseteq U_j$, and the second as U_i . By this we mean set $z_j = 1$ in the first, and $z_i = 1$ in the second. The transition function φ_{ij} is $\varphi_i \circ \varphi_j^{-1}$, which maps

$$\left(\left[\frac{z_0}{z_j} : \cdots : 1 : \cdots : \frac{z_n}{z_j} \right], t \right) \mapsto \left(\left[\frac{z_0}{z_j} : \cdots : 1 : \cdots : \frac{z_n}{z_j} \right], t[z] \right) = \left(\frac{z_j}{z_i}[z], t \frac{z_j}{z_i}[z] \right) \mapsto \left([z], \frac{z_j}{z_i} t \right).$$

Hence $g_{ij} = \frac{z_j}{z_i}$.

It will be useful to see how cocycles give rise to holomorphic vector bundles, since it is easier to see if cocycles are cohomologous in some sense, rather than constructing isomorphisms on the total space.

Suppose we are given, for all i, j , a cover $\{U_i\}_{i \in I}$ and maps $g_{ij} : U_{ij} \rightarrow \text{GL}(k, \mathbb{C})$ satisfying

- $g_{ij} = g_{ji}^{-1}$ and
- $g_{ij} \cdot g_{jk} = g_{ik}$.

Then we can build the total space and projection of a vector bundle of rank k $p : E \rightarrow X$ as follows. We start by taking the disjoint union of the covers, and then glue along the relation

$$E := \left(\coprod_{i \in I} U_i \times \mathbb{C}^k \right) / \langle (x, v) \sim (x, g_{ij}(v)) \forall x \in U_{ij} \rangle.$$

There is some bookkeeping needed to check E has a complex manifold structure. To define $p : E \rightarrow X$, choose $(x, v) \in [x, v]$, note that x is fixed, and map $p[x, v] = x$. Because the equivalence relation we glued along is linear, we find that all relevant universals for vector spaces can be done to vector bundles.

We compute the cocycles for the dual E^* for fun. Let $p : E \rightarrow X$ be a vector bundle of rank k with local trivializations $\varphi_i : p^{-1}(U_i) \rightarrow U_i \times \mathbb{C}^k$. Write $\varphi_i \circ \varphi_j^{-1} = g_{ij}$. When we dualize, the map g_{ij}^t goes the opposite direction, so we invert it to obtain $(g_{ij}^t)^{-1}$ as a cocycle for E^* . Proving this is a cocycle is like effort.

Let $E, F \rightarrow X$ be represented by $\{g_{ij}\}, \{h_{rs}\}$. WLOG assume they are trivialized on the same open cover, since otherwise we may refine it. We represent $E \oplus F$ by $g_{ij} \oplus h_{ij}$ (diagonal block matrix). We represent $E \otimes F$ by $g_{ij} \otimes h_{ij}$. We represent the *highest* exterior power $\Lambda^k E$, which should have rank 1, by $\{\det g_{ij}\}$.

Definition 1.1.54. Let $p : E \rightarrow X$ be a holomorphic vector bundle. A (global) holomorphic section of E is a holomorphic map $s : X \rightarrow E$ such that $p \circ s = 1$. We write $\Gamma(E)$ or $\Gamma(X, E)$ for the set of global sections of E .

One defines local sections in the same way. Note here that any holomorphic vector bundle E on X defines a sheaf on X , also denoted by E , by taking sections of the sheaf E to be sections of the vector bundle E .

If $\varphi_i : p^{-1}(U_i) \rightarrow U_i \times \mathbb{C}^k$ is a trivialization, then $\varphi_i \circ s|_{U_i} : U_i \rightarrow U_i \times \mathbb{C}^k$ is a vector-valued function $x \mapsto (x, (\varphi_i \circ s)(x))$. Here we abuse notation and write $\varphi(p) = (p, \varphi(p))$ since φ must act in this way. We also note that, writing $\varphi_i \circ s = f_i$, we have $f_i = g_{ij} f_j$.

Consider $\mathcal{O}_{\mathbb{CP}^1}(-1)$ and its dual $\mathcal{O}_{\mathbb{CP}^1}(1)$ on $\mathbb{CP}^1$. Represent these by g, h . Then

$$g_{01} = \frac{z_0}{z_1}, \quad h_{01} = \frac{z_1}{z_0} = \frac{1}{g_{01}}.$$

I got this backwards earlier. A section of $\mathcal{O}_{\mathbb{CP}^1}(-1)$ is a pair $f_0, f_1 : \mathbb{C} \rightarrow \mathbb{C}$ (which look like polynomials $a_0 + a_1t + \dots$ and $b_0 + b_1t^{-1} + b_2t^{-2} + \dots$), such that $f_0 = \frac{1}{t}f_1$ (agree under g_{ij} on overlap). There is no way for this to happen unless $f_0 = f_1 = 0$. Thus $\mathcal{O}_{\mathbb{P}^1}(-1)$ has no nontrivial sections, so $\Gamma(\mathbb{P}^1, \mathcal{O}_{\mathbb{P}^1}(-1)) \cong 0$. The same will not be true of the dual.

Lecture 10: Continuing holomorphic vector bundles

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We remark that $\mathcal{O}_{\mathbb{P}^n}(-1)$ is basically $\mathbb{P}^n \times \mathbb{C}^{n+1}$ apparently.

We then define

$$\mathcal{O}_{\mathbb{P}^n}(k) := \mathcal{O}_{\mathbb{P}^1}^{\otimes k},$$

and the corresponding cocycles are given by the tensors of the cocycles. Since the line bundles have dimension 1, the k -fold tensor of $g_{ij} = z_j/z_i$ is $g_{ij}^{\otimes k} = z_j^k/z_i^k$. Anyways the space of global sections $\Gamma(\mathbb{P}^n, \mathcal{O}_{\mathbb{P}^n}(k))$ is isomorphic to the homogeneous polynomials in $\{z_0, \dots, z_k\}$ of degree k . Recall of course that homogeneous polynomials are precisely polynomials in the coordinates of $\mathbb{P}^n$. In particular, the space is finite-dimensional.

Remark 1.1.55. We will see later, using Dolbeaux cohomology, that this generalizes: if $p : E \rightarrow X$ is a holomorphic vector bundle over a compact manifold has a finite dimensional $\mathbb{C}$ -vector space of global sections.

Example 1.1.56. Let X be a complex manifold with atlas $\{(U_i, \varphi_i)\}_{i \in I}$. Write $\psi_{ij} = \varphi_i \circ \varphi_j^{-1}$. Let $g_{ij} = J(\psi_{ij})$ for J the holomorphic Jacobian. This is a cocycle. Recall that taking the Jacobian preserves composition. Also note that $\psi_{ij} \circ \psi_{jk} = \psi_{ik}$, and thus $J(\psi_{ij} \circ \psi_{jk}) = J(\psi_{ij}) \cdot J(\psi_{jk}) = J(\psi_{ik})$. Additionally we have $J(f \circ f^{-1}) = J(1) = 1$ for any $f \in C^\infty(U)$, so the cocycle conditions are satisfied. We call the bundle this induces the holomorphic cotangent bundle $\mathcal{T}_X^*$, which will be the set of 1-forms $\Omega_{\text{holo}}^1(X)$. We will discuss the linear algebra of the total space construction later.

Remark 1.1.57. Let $\mathcal{L}_1, \mathcal{L}_2$ be line bundles on X . Then $\mathcal{L}_1 \otimes \mathcal{L}_2$ is also a line bundle. Additionally, $\Gamma(\mathcal{L}_1 \otimes \mathcal{L}_1^*, -) \cong \mathcal{O}_X$. To see this, if $\{g_{ij}\}$ represents $\mathcal{L}_1$ then $\{\frac{1}{g_{ij}}\}$ represents $\mathcal{L}_1^*$. Hence their product is 1. Alternatively, note that $\mathcal{L}_1 \otimes \mathcal{L}_1^*$ corresponds to endomorphisms of $\mathcal{L}_1$. Consider the identity and play with that.

Definition 1.1.58. Isomorphism classes of line bundles on a complex manifold X form a group of line bundles known as the *Picard group*, denoted $\text{Pic}(X)$.

We will see using more advanced machinery that $\text{Pic}(\mathbb{P}^n) \cong \mathbb{Z}^n$, a highly unusual fact following since $\mathbb{P}^n$ is some special type of variety.

Exercise 1.1.59. Let $p : E \rightarrow X$ be a holomorphic bundle. There is a correspondence between trivializations of a bundle $E|_U \rightarrow U \times \mathbb{C}^r$ and holomorphic frames on U .

Notation 1.1.60. Note that if $p : E \rightarrow X$ is any vector bundle and $x \in X$, then $p^{-1}(x) =: E_x$ is a vector space by choosing a trivialization.

Definition 1.1.61. Let $p_1 : E \rightarrow X$ and $p_2 : F \rightarrow X$ be holomorphic vector bundles on a complex manifold X . A homomorphism $\psi : E \rightarrow F$ is a holomorphic map such that

1. $\psi|_{E_x} : E_x \rightarrow F_x$ is linear, and
2. $\text{rk} \psi_x$ is constant as a function over $x \in X$, and

3. the diagram

$$\begin{array}{ccc}
 E & \xrightarrow{\psi} & F \\
 p_1 \searrow & & \swarrow p_2 \\
 & X &
 \end{array}$$

commutes.

We can construct kernels and cokernels of maps in the usual way, hence the category of holomorphic vector bundles is an abelian category.

We want to recast this definition in terms of cocycles. We start with trivial bundles. A homomorphism $\psi : E \rightarrow F$ gives us local maps $\psi_i : p^{-1}(U_i) \cong U_i \times \mathbb{C}^e \rightarrow p^{-1}(U_i) \cong U_i \times \mathbb{C}^f$ via $(x, v) \mapsto (x, \psi_i(x) \cdot v)$. Hence ψ_i is an $f \times e$ matrix of holomorphic functions. So we end up with a diagram

$$\begin{array}{ccc}
 p_1^{-1}(U_j) & \xrightarrow{\psi_j} & p_2^{-1}(U_j) \\
 g_{ij} \downarrow & & \downarrow h_{ij} \\
 p_1^{-1}(U_i) & \xrightarrow{\psi_i} & p_2^{-1}(U_i)
 \end{array}$$

Hence the general way to write a cocycle is to have matrices of holomorphic functions $\{\psi_i\}$ which are constant rank, and make this diagram commute.

Remark 1.1.62. Define E as a category with objects all U_i in the cover, and precisely one morphism $U_i \rightarrow U_j$ for all i, j . Hence this is a groupoid. We may then consider vector bundles as a functor, and morphisms of vector bundles are natural transformations between them. This is a somewhat naïve picture and there are indeed issues with trying to generalize this.

The correspondence of cocycles with holomorphic vector bundles is not precise, and is only bijective up to some equivalence relation on cocycles. We will explore this today, but the idea is that various open covers can admit the same cocycles.

Bibliography

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